

Objective : Invoking one REST API from another REST API

Refer - <https://restful-api.dev/>

Example - Get single object

To get details of object with given id

URL : <https://api.restful-api.dev/objects/{id}>

Objective - Continue with our blogs-api

Add a RestController , with GET to accept id

URL - http://localhost:8080/rest_clnt/{id}

This end point should call the - <https://restful-api.dev/>
n get details n send it to the client.

More 3rd party APIs

<https://apiclub.readme.io/reference/ifsc-code-lookup-api>

How to make a REST call from one web app to another ?

Use : `org.springframework.web.client.RestTemplate`

Other alternative : `WebClient`

The `RestTemplate` class in Spring Framework is a synchronous HTTP client for making HTTP requests to consume RESTful web services.

It exposes a simple and easy-to-use template method API for sending an HTTP request and also handling the HTTP response.

The `RestTemplate` class also provides aliases for all supported HTTP request methods, such as GET, POST, PUT, PATCH , DELETE, and OPTIONS.

Steps -

1. Configure `RestTemplate` as a spring bean (`@Bean`) in the Spring boot application class.

`@Bean`

```
public RestTemplate restTemplate(){  
    return new RestTemplate();  
}
```

2. Inject it as a dependency in the service Layer

`@Autowired`

```
private RestTemplate restTemplate;
```

3. Instead of hard coding the URLs , add them in application.properties file

REST Client URL

```
get.url=https://api.restful-api.dev/objects/{id}
```

4. Using SpEL , inject it in the service layer

`@Value("${get.url}")`

```
private String getURL;
```

5. Make REST API call using methods of the `RestTemplate` class.

```
public <T> T getForObject(String url,Class<T> responseType,Object... uriVariables)
```

throws RestClientException
T - type of the response